

Khaja Mujahiddin Mohammed

Senior Software Engineer | Backend & Data Platforms

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PROFESSIONAL SUMMARY

Senior Software Engineer with mid level of experience designing and operating high-throughput microservices, distributed pipelines, and cloud backend platforms. Proven track record of end-to-end service ownership, from architectural design and SOLID object-oriented implementation to CI/CD, observability, and production support. Core expertise in Python, Java, TypeScript, SQL/NoSQL databases, and event-driven architectures. Expert at leading code reviews, driving code quality standards, and scaling mission-critical platforms to **99.99% availability**.

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Data Platform

July 2025 – Present

Optum (UnitedHealth Group)

Hartford, CT

- Owned design and deployment of distributed ETL and microservice platform components using Python, FastAPI, and PySpark on AWS/Azure, improving throughput by **35%**.
- Architected resilient REST APIs and async messaging pipelines with Kafka and SQS, cutting downstream API latency to **sub-100ms** at peak loads.
- Implemented rigorous CI/CD workflows using GitHub Actions and Docker, reducing integration bugs by **25%** and achieving zero-downtime releases.
- Established comprehensive logging and observability with CloudWatch and Datadog, decreasing Mean Time to Resolution (MTTR) by **40%**.
- Mentored 4 junior engineers, led weekly code reviews, and enforced SOLID object-oriented standards across cross-functional teams.

Software Engineer

January 2022 – July 2023

Staples

Chennai, India

- Re-architected core B2B payment gateway microservices using Java 17, Spring Boot, and PostgreSQL, lifting authorization success from 97% to **99.7%**.
- Optimized relational database queries, indexing strategies, and multi-tier Redis caching, slashing p95 reporting API response times from **800ms to 600ms**.
- Engineered near-real-time data synchronization between MongoDB order datastores and PostgreSQL warehouses across **40+ regional markets**.
- Built robust unit and integration test suites in JUnit and PyTest, maintaining **90%+ code coverage** across all production services.
- Participated in 24/7 on-call rotation for high-throughput transactional services, maintaining **99.9% uptime** during Black Friday peaks.

Software Engineer

August 2019 – December 2021

SRJK Consulting (Client: BMW Group)

Hyderabad, India

- Developed async vehicle telemetry ingestion services for **50,000+ connected vehicles** using Java, gRPC, and Kafka Streams.
- Designed partitioned PostgreSQL and Redis schemas for 100M+ configuration records, accelerating dealer feature onboarding by **50%**.
- Containerized enterprise legacy monolithic monolith into Docker microservices deployed on AWS EKS, increasing release frequency **3x**.

- Constructed backward-compatible gRPC and Protobuf API contracts, reducing cross-team platform integration times from **3 weeks to 5 days**.
- Authored architectural decision records (ADRs), runbooks, and wiki operational documentation for **15+ distributed microservices**.

EDUCATION

M.S., Data Science

University of New Haven

August 2023 – May 2025

New Haven, CT

Coursework: advanced ML, deep learning, NLP, cloud MLOps, big data. Key projects include multimodal misinformation detection (~85% accuracy) and U-Net pedestrian/cyclist segmentation.

B.Tech

Sreenidhi Institute of Science and Technology

June 2017 – May 2021

Hyderabad, India

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C#, SQL, Go

Backend & Web: Node.js, FastAPI, Spring Boot, REST API, GraphQL, gRPC, Microservices, SOLID Principles

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Snowflake, SQL Server, NoSQL

Data & Event Streaming: Apache Kafka, AWS SQS, PySpark, dbt, Airflow, ETL/ELT

Cloud & DevOps: AWS (S3, Lambda, ECS, EKS), Azure, Docker, Kubernetes, CI/CD, Terraform, GitHub Actions

Testing & Observability: PyTest, JUnit, Prometheus, Grafana, Datadog, ELK Stack, OpenTelemetry

PROJECTS

VERITAS HITL Platform

github.com/khajamujahid/veritas-hitl-platform

Multimodal Fake News Detection

- Human-in-the-loop multimodal platform: Gemini extracts verifiable claims (strict Pydantic schema), then Google Fact Check API results feed a forensic report for reviewers.
- Image/video forensics: ELA heatmaps, EXIF flagging, reverse image search; video keyframes, optical flow, per-frame ELA.
- Celery + Redis offload heavy jobs so the API stays responsive; Docker Compose for web + Redis + workers; React/Tailwind dashboard.

MIKA OS / Clinical AI Agent

github.com/khajamujahid/Clinical-AI-Agent

Full-Stack Clinical RAG Companion

- RAG healthcare companion: FastAPI + LangChain retrieves from ChromaDB (HuggingFace embeddings) before Gemini generation over synthetic patient data.
- Hallucination guardrails constrain answers to retrieved context; Next.js 3D clinical UI for natural-language exploration.

Pedestrian & Cyclist Segmentation (U-Net)

Computer vision

- Pixel-level segmentation of pedestrians and cyclists in urban scenes using a U-Net architecture — CV engineering for perception-style tasks.

CERTIFICATIONS

Oracle Cloud Infrastructure — Multicloud Architect Associate

Oracle Analytics Cloud (OAC)